

Md. Mahmudul Hasan

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SUMMARY

Backend developer with hands-on experience building production APIs and real-world systems. Passionate about software architecture and systems design I build things from scratch to understand how they actually work at a deeper level, with a habit of going beyond the surface of the tools I use.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C++, Java
Backend Frameworks: Django, FastAPI (familiar) Express.js, Node.js, NestJS
Databases & Caching: PostgreSQL (Prisma), MongoDB (Mongoose), Redis, SQLite
Core Skills: RESTful API Design, OOP, Data Structures & Algorithms, Problem Solving
Developer Tools: Git, GitHub, Docker (familiar), VS Code, Vercel, Firebase
Frontend: React.js, Next.js, HTML, CSS, SCSS

EXPERIENCE

Jr. Backend Developer

June 2025 – October 2025

Backbencher Studio

- Built and optimized RESTful APIs with Express.js and Node.js, implementing JWT-based authentication and resolving production issues to improve system reliability.
- Contributed to backend integrations for filtering, order management, AI integration and data pipelines.
- Collaborated with team using Git and GitHub for version control, code review, and feature branching.

PROJECTS

MahmudCore | *Python, WSGI, Jinja2, WebOb, PyPI* | [PyPI](#) | [GitHub](#)

- Built a WSGI-compliant Python web framework from scratch, implementing core internals including URL routing with dynamic parameters, class-based and function-based request handlers, and HTTP method control.
- Engineered a middleware pipeline architecture, Jinja2 template engine integration, custom response helpers (JSON/HTML/text), static file serving via WhiteNoise, and a global exception handling system.
- Packaged and published the framework to PyPI with full test coverage using pytest, a built-in WSGI test client, and semantic versioning - making it installable via `pip install MahmudCore`.

Redis-Inspired In-Memory Store | *Python, TCP, Socket, Event Loop* | [GitHub](#)

- Built a Redis-inspired in-memory key-value store from scratch in Python, implementing a single-threaded event loop with non-blocking I/O via `select()` to handle multiple concurrent clients without locks or race conditions.
- Implemented TTL-based key expiration using both lazy (on-access) and active (background probabilistic) cleanup strategies, along with support for core Redis commands including `SET`, `GET`, `EXPIRE`, `TTL`, and `KEYS`.
- Engineered dual persistence mechanisms - AOF (Append-Only File with fsync policies and background rewrite) and RDB (binary snapshot with background save) - enabling crash-safe recovery and data durability.

PHARMA Web Application | *Redux, Next.js, TypeScript, Node.js, Express.js, MongoDB, Mongoose* | [GitHub](#)

- Built a full-stack medicine management system with Next.js, TypeScript, role-based access, and admin controls.
- Developed secure RESTful APIs with JWT authentication, role-based authorization, and full medicine CRUD operations.
- Implemented search, filtering, and order management features with optimized Mongoose models for seamless API integration.

EDUCATION

National Institute of Textile Engineering and Research

Bachelor of Science in Computer Science and Engineering — CGPA: 3.60/4.00

Dhaka-1350, Bangladesh

Feb. 2022 – Jun. 2026

CERTIFICATIONS AND ACHIEVEMENTS

Supervised Machine Learning: Regression and Classification — Coursera (Andrew Ng)

Next Level Web Development — Programming Hero

1st Place, Intra Research Fair Competition - 2026 — CRIR (NITER)

NextGen Hakathon - 2025 — Participant (IIUC)